Computer Vision Project Assignment (5th Semester)

There are three project assignments listed below. Each student is required to select **one project** and submit it by **20th October** as part of the mid-term evaluation.

These assignments will form an important component of your assessment. If, for any reason, you anticipate difficulty in completing the chosen assignment, please reach out to your faculty at the earliest for guidance or support.

Assignment 1: Smart Society Security System (Face Recognition)

Objective: Build a security system that uses computer vision to identify individuals entering a premises.

- **Description:**
  - Use a web camera (use laptop webcam) feed to capture faces of people entering.
  - Compare captured faces against a pre-registered database of residents.
  - Allow entry if the person is recognised; otherwise, flag or reject entry. Show a message on the screen directly (Entry Allowed in Green, Entry Rejected in RED)
- **Tech Stack:** OpenCV, Python, face recognition libraries (dlib/FaceNet), SQLite/MySQL for database.
- **Deliverables:**
  - Working prototype with laptop webcam feed.
  - Database of at least 20 registered faces.
  - Report on accuracy, false positives/negatives.

Assignment 2: Recognise license plates using OCR

Objective: Detect vehicles violating traffic rules using computer vision.

- **Description:**
  - Use video footage of a vehicles at traffic junction.
  - Recognise license plates using OCR (Tesseract or EasyOCR).
- **Tech Stack:** OpenCV, YOLO/SSD for object detection, Python, OCR library.
- **Deliverables:**
  - Demo video showing detection of violations.
  - Accuracy report on detection.

Assignment 3: AI-Powered Attendance System

Objective: Automate classroom attendance using face recognition.

- **Description:**
  - Camera scans the classroom at the start of a lecture.
  - Recognises students' faces against a pre-registered database.
  - Marks attendance automatically in a CSV/DB file.
- **Tech Stack:** OpenCV, Python, face recognition library, Pandas/SQLite for attendance logs.
- **Deliverables:**
  - Prototype system with at least 10 registered students.
  - Attendance log file generated automatically.
  - Evaluation of system accuracy and limitations.